

Wireless Tape Measurer

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TECHIN 514

Final Project

Github Repo: <https://github.com/vpon-lab/514-Final-Project>



Measuring alone

Problem to Solve:

- > **Wanting to measure bedroom space on your own with out another person's assistance.**
- > **Mitigate sag from a physical tape measurer.**

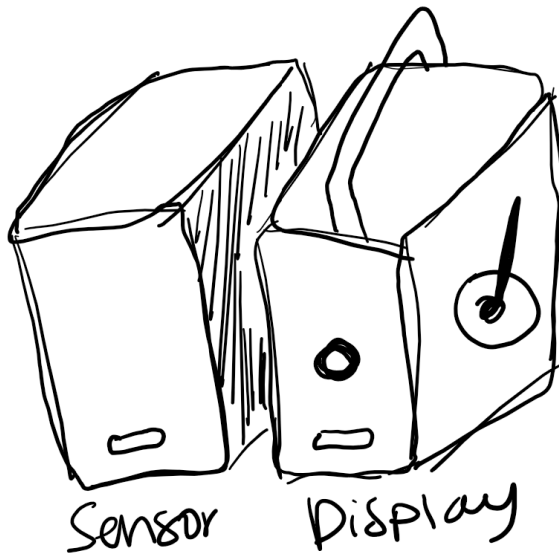


Measuring alone

A proposed Solution:

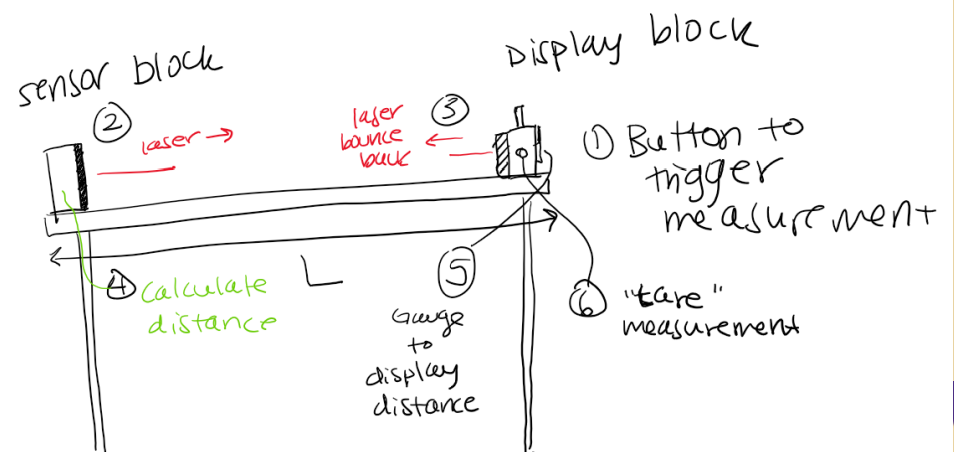
- > **Wireless tape measurement**
- > **2-piece system to measure from point A to point B**

2-Block System

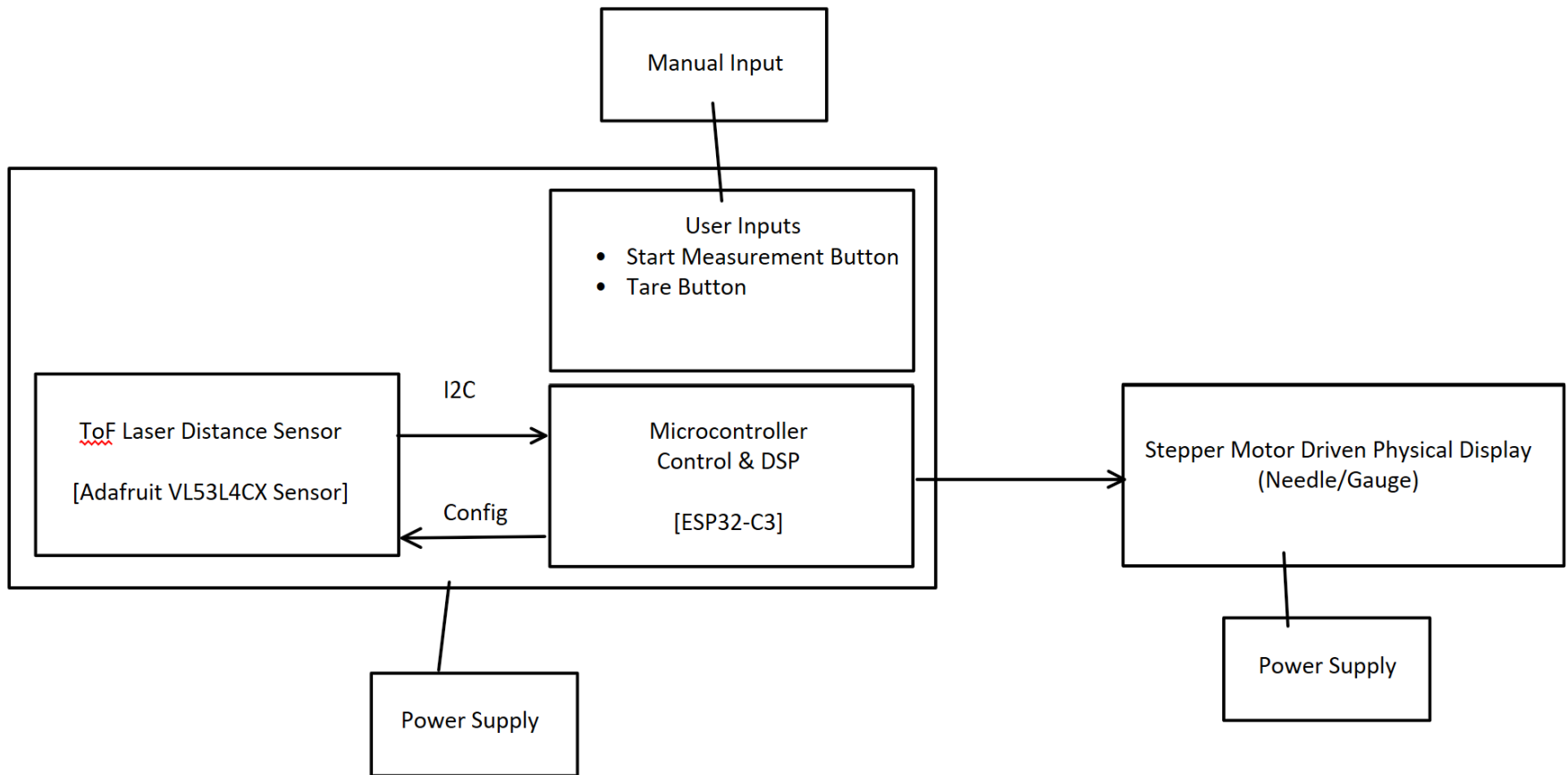


Communication and Diagram

Scenario: measuring table length 'L'



System Architecture



Sensing + Display Device

Due to multiple ESP32C3 Battery pad mishaps, this is not battery powered for cautionary measures



Sensor Device

XIAO
ESP32C3

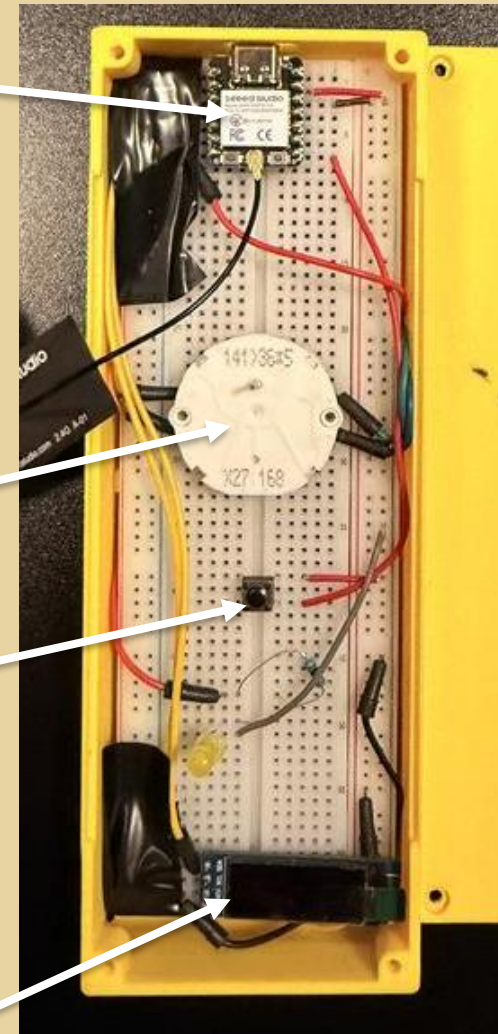
Time of
Flight Laser
Sensor

X27 Stepper
Motor

"Tare"
Button

Power
On/Off
Switch

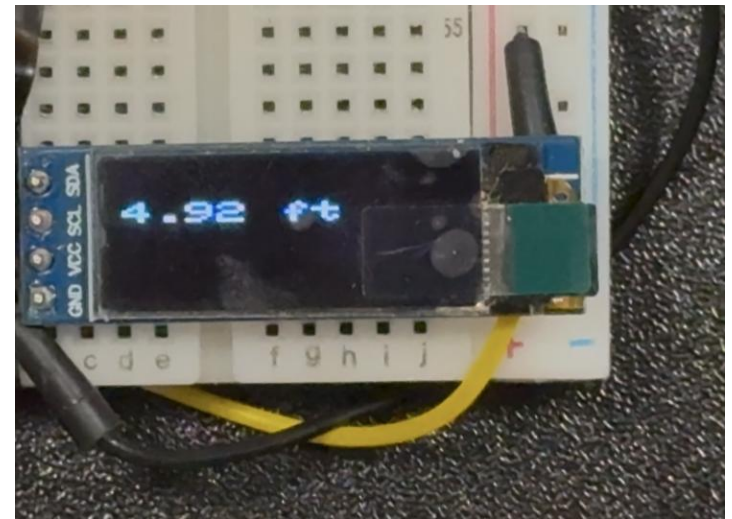
OLED
Display



Display Device

Signal Processing

- > Moving average filter, numbers were the same at the first decimal place before filter.
- > Accuracy:
 - ± 0.01 ft
 - ± 3 mm
 - Degrades with distance

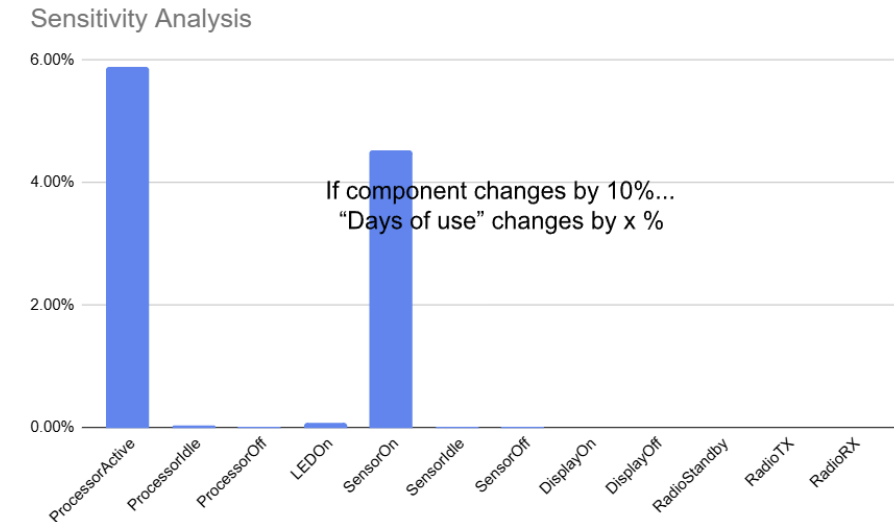


Display Device shows sensor jumping between measurements – from table to ceiling



Battery Considerations – Sensing device

- > 3.7V Lipo or Li-Ion Battery
- > Compact easy to design with in enclosure and long lasting
- > Based on 2hrs/day of use



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Appendix

Anything with “\$0” was
free in the Maker Space

Budget Summary

Supplies		
Purchased	Qty	Total Price
XIAO ESP32C3	2	0
ST VL53L4CX	2	
3.7V Li-Ion Batteries	2	\$14.96
Battery holders	2	\$5.99
X27 Stepper Motor	1	0
3.6 kOhm Resistors	2	0
220 Ohm Resistor	1	0
Red LED	1	0
OLED SSD1306	1	0
Tactile Switch	1	0
Single-Pole, Double-Throw (SPDT) slide switch	2	0
3M Reflective Tape	1	\$9.05
Plexiglass Clear Acrylic Sheets Panel	1	\$9.99
Total		39.99

Future Work

- > Sensor Calibration to get rid of cross-talk from clear acrylic window.**
- > Sensor Calibration to find max capable distance based on Bluetooth connection compatibility**
- > Develop a more ergonomic and detachable/portable enclosure(s)**
- > Proper PCB design for easy battery connection to Display device.**

Appendix

Power Model (Sensing Device)

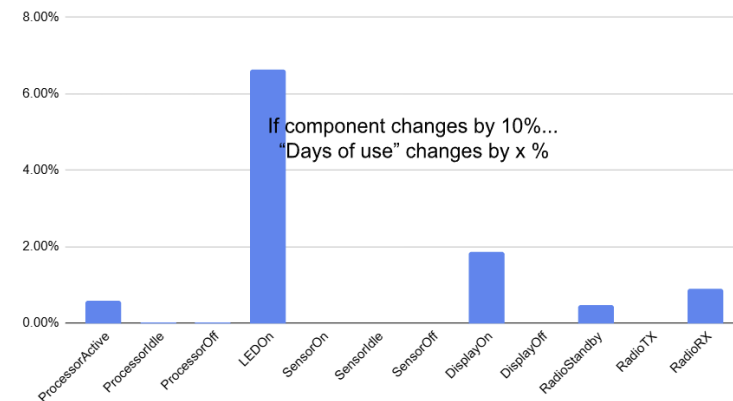
System Parameters (defined by hardware)			Profiles (usage of each component mode - defined by software and usage)		
	from the datasheets				
Supply (V)	3.3		"off"	"sensing"	"interactive"
Processor			Current Consumption (mA)		
Active (modem sleep)	89 mW		27	0%	10%
Idle	0 mW		0.13	0%	85%
Sleep	0 mW		0.005	100%	5%
LED					
On	1 mW		0.17	0%	100%
Sensor					
On	69 mW		21	0%	100%
Idle	0 mW		0.009	0%	0%
Off	0 mW		0.007	100%	0%
Display					
On	0 mW		0	0%	0%
Off (leakage)	0 mW		0	0%	0%
Radio					
Data Rate	400 bps				
Standby Power	3 mW				
TX Power	85 mW				
RX Power	85 mW				
			20	2	2 hours/day typical usage
Battery					
Capacity	1892 mAh				
Nominal Voltage	4 V				
Regulator Efficiency	85%				

Appendix

Power Model (Display Device) if it were battery powered

System Parameters (defined by hardware)		Profiles (usage of each component mode - defined by software and usage)		
form the datasheets		"off"	"sensing"	"interactive"
Processor				
Active	76 mW	0%	80%	50%
Idle	0 mW	0%	20%	50%
Sleep	0 mW	100%	0%	0%
LED				
On	48 mW	100%		0%
Sensor				
On	0 mW	0%	0%	0%
Idle	0 mW	0%	0%	0%
Off	0 mW	0%	0%	0%
Display				
On	155 mW	0%	100%	100%
Off (leakage)	0 mW	100%	0%	0%
Radio				
Data Rate	1000000 bps	0%	0%	0%
Standby Power	56 mW	0%	95%	50%
TX Power	1106 mW	0%	0%	0%
RX Power	277 mW	0%	5%	50%
Battery		22	1	1 hours/day typical usage
Capacity	2500 mAh			
Nominal Voltage	4 V			
Regulator Efficiency	85%			

Sensitivity Analysis



Total power in profile (mw)		Maximum Time	
"off"	48.0165 mW	163.7	hours
"sensing"	283.2608 mW	27.8	hours
"interactive"	360.1145 mW	21.8	hours
Effective Battery Capacity			
		7862.5	mW*h
Days of Use	4.63 days		
Hours of Use	111.02 hours		

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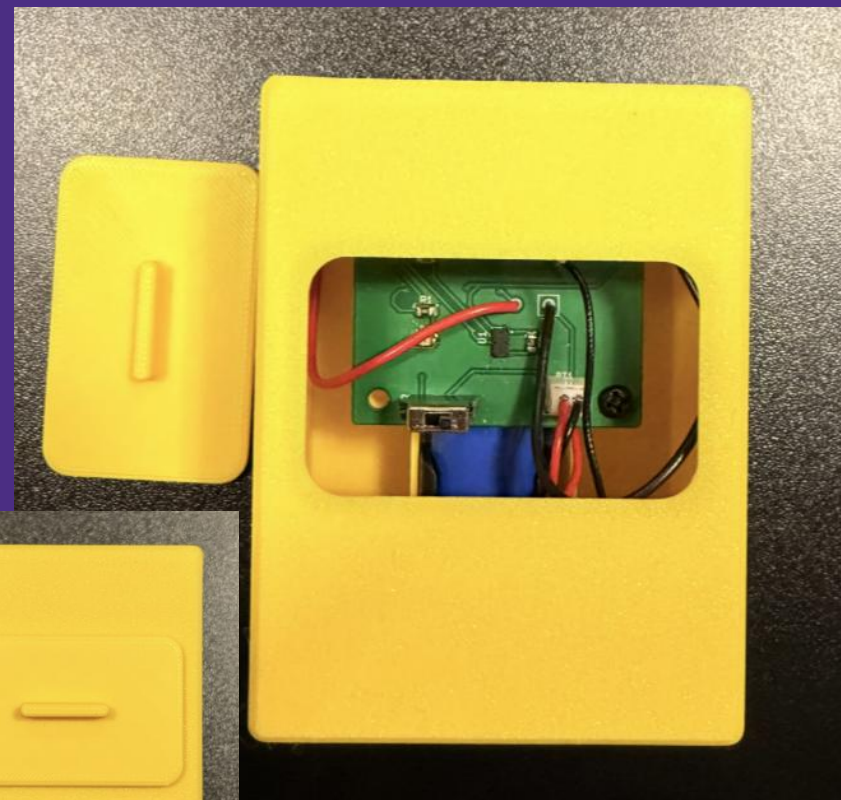
Appendix

Enclosure Design



Display Device

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Sensor Device